

Sandip Singha

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SUMMARY

Backend-Focused Full-Stack Developer with expertise in architecting high-throughput REST APIs, distributed microservices, and asynchronous message-processing pipelines. Proficient in Node.js, Express.js, PostgreSQL, MongoDB, and Redis, with proven experience implementing LLM integrations, real-time Socket.io architectures, and resilient queue-driven workflows. Adept at building modern web interfaces using React.js and Next.js to deliver end-to-end applications.

WORK EXPERIENCE

Fullstack Developer, Indomitech Global Pvt. Ltd. (formerly Indomitech Group) *Apr 2025 – Present*

- Engineered and maintained **REST APIs** using Node.js and Express.js for mobile and web clients.
- Developed responsive frontend interfaces using **React.js and Next.js** for cross-platform applications.
- Integrated **Firebase Push Notifications and deep linking** for Android and iOS applications.
- Integrated **Google Maps API with Socket.io** for real-time tracking, location sharing, and distance calculation.
- Leveraged **OpenAI APIs** to generate automated Asset Management Reports and streamline reporting workflows.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Python, Java, C, PHP, React.js, Next.js
Backend & Databases: Node.js, Express.js, REST APIs, Socket.io, MongoDB, PostgreSQL, Drizzle ORM, Redis
DevOps & Others: BullMQ, RabbitMQ, Docker, Git, GitHub Actions, AWS

PROJECTS

AskPDF — RAG-Powered Document Q&A System [GitHub](#)

Node.js, Express.js, Next.js, LangChain, Qdrant, MongoDB, Redis, BullMQ, Gemini API

- Built a **RAG-based document Q&A system** enabling PDF upload and conversational querying with document-grounded responses and **real-time streaming**.
- Designed a **5-stage asynchronous pipeline** (parse → chunk → embed → index → search) using LangChain and Qdrant for semantic retrieval.
- Decoupled PDF processing through a **BullMQ worker backed by Redis**, returning **HTTP 202 Accepted** immediately while tracking processing status in MongoDB.
- Implemented **document-scoped retrieval and similarity thresholding** (≥ 0.7) to filter low-relevance context before Gemini generation, with **SSE streaming** and source page citations.

Real-Time Chat Application [GitHub](#)

Next.js, Express.js, Socket.io, RabbitMQ, Redis, AWS

- Architected a **microservice-based chat platform** with low-latency bidirectional communication using Socket.io.
- Implemented reliable **message queuing with RabbitMQ** and Redis caching for performance under high concurrency.
- Implemented **Redis caching** for fast data access and optimized real-time updates across connected clients.

Public News & Media Platform *Company Project — Indomitech Global Pvt. Ltd.*

- Developed frontend using **Next.js** and backend APIs using **Node.js and Express.js** for a public news platform.
- Implemented **role-based access control (RBAC)** and administrator workflows for managing users, content, and platform permissions.
- Built **scheduled post publishing** workflows and integrated **Razorpay payment gateway** for subscription/payment processing.
- Implemented server-side **push notifications and deep linking** to support Android and iOS applications.

EDUCATION

2025 – 2027 **Master of Computer Applications**, JIS College of Engineering *Pursuing*
2022 – 2025 **Bachelor of Computer Applications**, JIS College of Engineering *CGPA: 7.90*